Algebraic Metrization and Native Arithmetic over Onion de Bruijn Sequences

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Abstract

We establish an exact closed-form algebraic bijection mapping the infinite Onion de Bruijn sequence to absolute integer positions for dimensions $n = 2$ and $n = 3$. We utilize this algebraic structure to define a native *Dynamic Block-Carry* string arithmetic, allowing for $\mathcal{O}(1)$ sequence addition $w_1 \oplus w_2$. Finally, we extend this Arithmetic Logic Unit (ALU) for arbitrary $n \geq 4$ by reducing coordinate projections strictly to polynomial-time $\mathcal{O}(n^2 \log k)$ combinatorial FKM ranking and unranking bounded within the layer graph $\mathrm{Lay}(n, k)$.

1 Introduction and Definitions

Definition 1.1. *1. Let $\mathrm{DB}(n, k)$ with $n, k \in \mathbb{N}$ denote the standard directed de Bruijn graph whose vertices are words of length n over an alphabet $[k] = \{0, 1, \ldots, k-1\}$, equipped with standard left-shift adjacency (a directed edge from u to v exists if $u = \sigma x$ and $v = x\tau$ for some word $x \in [k]^{n-1}$ and characters $\sigma, \tau \in [k]$).*

2. We define the (n, k)-layer of the de Bruijn graph, denoted by $\mathrm{Lay}(n, k)$, as the induced subgraph of $\mathrm{DB}(n, k)$ consisting of all vertices in $[k]^n \setminus [k-1]^n$. Consequently, every word natively in $\mathrm{Lay}(n, k)$ contains the maximal character $k - 1$.

3. An (n, k)-layer de Bruijn sequence is defined as a Hamiltonian cycle on $\mathrm{Lay}(n, k)$.

4. An (n, k)-Onion de Bruijn sequence is a cycle cover of the vertices of $\mathrm{DB}(n, k)$ consisting of k disjoint cycles, such that each cycle represents an independent Hamiltonian traversal on the specific layer $\mathrm{Lay}(n, j)$ for $1 \leq j \leq k$.

5. A sequence metric $p_\infty(w)$ is a bijective mapping from a word $w \in \mathbb{N}^n$ to its absolute sequence position in the infinite Onion sequence. This defines arithmetic operations natively over the sequence states:

$$w_1 \oplus w_2 = p_\infty^{-1}\Big(p_\infty(w_1) + p_\infty(w_2)\Big).$$

Notation 1.2.

- $|\,\mathrm{DB}(n, k)|$ *denotes the total number of (n, k)-de Bruijn sequences.*
- $|\,\mathrm{Lay}(n, k)|$ *denotes the total number of (n, k)-layer de Bruijn sequences.*

2 Algebraic Metrization

Because the infinite Onion sequence constructs layers chronologically, the base integer volume preceding layer k is exactly the volume of all prior layers combined: $V_k = (k-1)^n$. The remaining local offset is strictly bounded by the volume of the current layer, $k^n - (k-1)^n$. Thus, any word's absolute integer position is uniquely determined by $p_\infty(w) = (k-1)^n + L(w)$, where $L(w)$ is the local positional offset inside the layer graph $\mathrm{Lay}(n,k)$, bounded strictly by $0 \le L(w) < k^n - (k-1)^n$.

2.1 Exact Native Metrization for $n = 2$ and $n = 3$

For $n \le 3$, prime length divisors guarantee uniform cycle structures on $\mathrm{Lay}(n,k)$. Because the infinite sequence builds chronologically, the piecewise coordinate rules evaluating the local offset $L(w)$ natively map to the layer parameter k.

Definition 2.1 (Native Coordinate Offset for $n = 2$)**.** *For any word $w = (x,y) \in \mathrm{Lay}(2,k)$, its maximal character is strictly $k-1$. The local positional offset $L_2(x,y)$ inside the layer is evaluated strictly algebraically:*

$$L_2(x,y) = \begin{cases} 2y, & \text{if } x = k-1 \text{ and } y < k-1, \\ 2x-1, & \text{if } y = k-1 \text{ and } 0 < x < k-1, \\ 2k-3, & \text{if } x = k-1 \text{ and } y = k-1 \text{ and } k > 1, \\ 2k-2, & \text{if } x = 0 \text{ and } y = k-1. \end{cases} \tag{1}$$

Conversely, any valid local offset $L \in [0, 2k-2]$ within layer k can be projected directly back to its physical string coordinates natively via the inverse projection $L_2^{-1}(L,k)$:

$$L_2^{-1}(L,k) = \begin{cases} (k-1, L/2), & \text{if } L < 2k-3 \text{ and } L \text{ is even}, \\ ((L+1)/2, k-1), & \text{if } L < 2k-3 \text{ and } L \text{ is odd}, \\ (k-1, k-1), & \text{if } L = 2k-3 \text{ and } k > 1, \\ (0, k-1), & \text{if } L = 2k-2. \end{cases} \tag{2}$$

Theorem 2.2 (Native Coordinate Offset for $n = 3$)**.** *For any word $w = (a,b,c) \in \mathrm{Lay}(3,k)$, we identify its canonical rotation $w^* = (k-1, x, y)$ defined as the lexicographically maximal cyclic shift of w. The Prefer-Max sequence positional offset $f_3(w)$ is evaluated as follows: If $w = (0,0,k-1)$, then $f_3(w) = 0$. Otherwise, let the internal subset offset $g(x)$ be defined as:*

$$g(x) = \begin{cases} 1, & \text{if } x = k-1, \\ 3(k-1)^2 - 3(k-1)x + 2, & \text{if } x < k-1. \end{cases} \tag{3}$$

If $w = (0, k-1, x)$, then $f_3(w) = g(x)$. For all other words, $f_3(w) = g(x) + s(x,y) + c(w) + 1$, where the internal sub-cycle offset $s(x,y)$ is:

$$s(x,y) = \begin{cases} 0, & \text{if } x = k-1 \text{ and } y = k-1, \\ 3(k-1-y) - 2, & \text{if } x = k-1 \text{ and } y < k-1, \\ 3(k-1-y) - 3, & \text{if } x < k-1, \end{cases} \tag{4}$$

and $c(w)$ represents the discrete cyclic trace offset relative to w^*. For $y > 0$:

$$c(w) = \begin{cases} 0, & \text{if } w = (k-1, x, y), \\ 1, & \text{if } w = (x, y, k-1), \\ 2, & \text{if } w = (y, k-1, x). \end{cases} \tag{5}$$

For $y = 0$, the trace restricts exclusively to the remaining cyclical shifts: $c(w) = 0$ if $w = (k-1, x, 0)$ and 1 if $w = (x, 0, k-1)$.

Because the infinite Onion sequence operates chronologically from the origin, the local offset $L_3(w)$ is mathematically the direct reverse of the Prefer-Max offset:

$$L_3(w) = 3k^2 - 3k - f_3(w). \tag{6}$$

Proof. Because $n = 3$ is prime, periodic Lyndon degeneracies are strictly limited to uniform fixed points of length 1. Summing the volumes of these subsets exactly yields $k^3 - (k-1)^3 = 3k^2 - 3k + 1$, matching the exact total combinatorial volume of $\mathrm{Lay}(3, k)$. The piecewise constraints strictly bound these rotational subsets, preserving geometric closure without fractional periodicity. □

3 Native String Arithmetic: Dynamic Block-Carry

In standard base-10 arithmetic, a "carry" is triggered when a column sum exceeds 9. By mapping the bijection $p_\infty(w)$ directly to the sequence coordinates, we define a native Arithmetic Logic Unit (ALU). In the Onion de Bruijn topology, a native string-carry is triggered when the algebraic sum overflows the geometric capacity of the current layer graph $\mathrm{Lay}(n, k)$.

We define native sequence addition $w_{\text{new}} = w_1 \oplus w_2$ directly on the coordinates through the following ALU:

1. **Extract Topological Volumes:** Identify layers $k_1 = \max(w_1) + 1$ and $k_2 = \max(w_2) + 1$. Evaluate their local offsets L_1 and L_2 purely via native coordinate rules.

2. **Summation:** Compute the total combined integer volume $S = ((k_1 - 1)^n + L_1) + ((k_2 - 1)^n + L_2)$.

3. **Geometric Block-Carry:** Extract the destination layer by taking the discrete n-th root of the combined volume:

$$k_{\text{new}} = \lfloor \sqrt[n]{S} \rfloor + 1. \tag{7}$$

   The remainder determines the new local offset inside the destination layer $\mathrm{Lay}(n, k_{\text{new}})$:

$$L_{\text{new}} = S - (k_{\text{new}} - 1)^n. \tag{8}$$

4. **Symbol Projection:** Map L_{new} directly back to the new string coordinates using the inverse piecewise projection bounded by k_{new}.

Worked Example: Tracing $(0, 2) \oplus (1, 0) \mapsto (3, 0)$ We trace the example to evaluate the operation without iteratively generating the sequence.

- Evaluate $w_1 = (0, 2) \in \mathrm{Lay}(2, 3)$. Base volume $(3-1)^2 = 4$. Local offset $L_1(0, 2) = 2(3) - 2 = 4$. Total w_1 volume $= 8$.

- Evaluate $w_2 = (1, 0) \in \mathrm{Lay}(2, 2)$. Base volume $(2-1)^2 = 1$. Local offset $L_2(1, 0) = 0$. Total w_2 volume $= 1$.

- Geometric Carry: $S = 8 + 1 = 9$. New layer $k_{\text{new}} = \lfloor \sqrt{9} \rfloor + 1 = 4$. New offset $L_{\text{new}} = 9 - (4-1)^2 = 0$.

- Symbol Projection: $L_2^{-1}(0,4) \implies L < 2(4) - 3$, even. $x = 4 - 1 = 3,\ y = 0/2 = 0 \implies (\mathbf{3}, \mathbf{0})$.

Thus, we deduce natively on the coordinates that the volume of the second state pushes the first state exactly into the root of the next topological layer.

4 Generalized ALU via Layer-Restricted FKM Ranking

The elegance of the $n \leq 3$ native string arithmetic relies on prime length divisors guaranteeing uniform cycle structures over the layer graphs. For $n \geq 4$, internal periodicities (such as $w = (k-1, x, k-1, x)$) induce sub-cycles of varying lengths, breaking the symmetry required for simple piecewise evaluation.

To evaluate distances algebraically for arbitrary $n \geq 4$, we restrict the $\mathcal{O}(n^2 \log k)$ combinatorial Lyndon ranking and unranking strictly to the layer graph $\mathrm{Lay}(n, k)$.

Theorem 4.1. *For arbitrary $n \geq 4$, the local offset $L_n(w)$ inside the infinite topological layer $\mathrm{Lay}(n, k)$ reduces in strictly polynomial time to the combinatorial FKM rank of the complemented word restricted to the layer graph.*

Proof. Because the finite Prefer-Max greedy algorithm exactly exhausts all words containing the maximal character $k-1$ before visiting any words restricted strictly to $[k-1]^n$, the initial prefix of the finite Prefer-Max sequence over $[k]^n$ perfectly traces the layer graph $\mathrm{Lay}(n, k)$.

Because the infinite Onion sequence appends these layers sequentially by reversing the order of the Prefer-Max traversal, the local offset $L_n(w)$ inside the infinite sequence layer is the direct reverse of its rank in the finite Prefer-Max sequence:

$$L_n(w) = k^n - (k-1)^n - 1 - R_{\mathrm{Pmax}}(w). \tag{9}$$

By the FKM Theorem [2], the Prefer-Min sequence is the exact lexicographical concatenation of Lyndon words. Prefer-Max operates as the symbol-wise complement, forming a cyclic shift of the complemented FKM sequence by exactly $n-1$ positions. Thus, the rank $R_{\mathrm{Pmax}}(w)$ evaluates algebraically:

$$R_{\mathrm{Pmax}}(w) = \left(R_{\mathrm{FKM}}(\overline{w}^{(k-1)}, k) + n - 1 \right) \bmod k^n, \tag{10}$$

where $\overline{w}^{(k-1)}$ is the complement defined by $\sigma_i \mapsto (k-1) - \sigma_i$, and R_{FKM} is the rank index within the FKM sequence over alphabet size k.

Since Kociumaka et al. [1] established that R_{FKM} evaluates natively in $\mathcal{O}(n^2 \log k)$ time via Lyndon word ranking, we extract $L_n(w)$ polynomially without Eulerian state traversal. The Arithmetic Logic Unit is fully generalized to arbitrary n, computing distance natively as operations on the layer graphs. □

Code Availability

A complete Python implementation of the operations and arithmetic logic described in this paper is publicly available on GitHub at https://github.com/DaftCode101/Onion-De-Bruijn/blob/main/prefer-max-operations.py.

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References

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[2] H. Fredricksen and J. Maiorana. (1978). *Necklaces of beads in k colors and k-ary de Bruijn sequences*. Discrete Mathematics, 23(3):207–210.